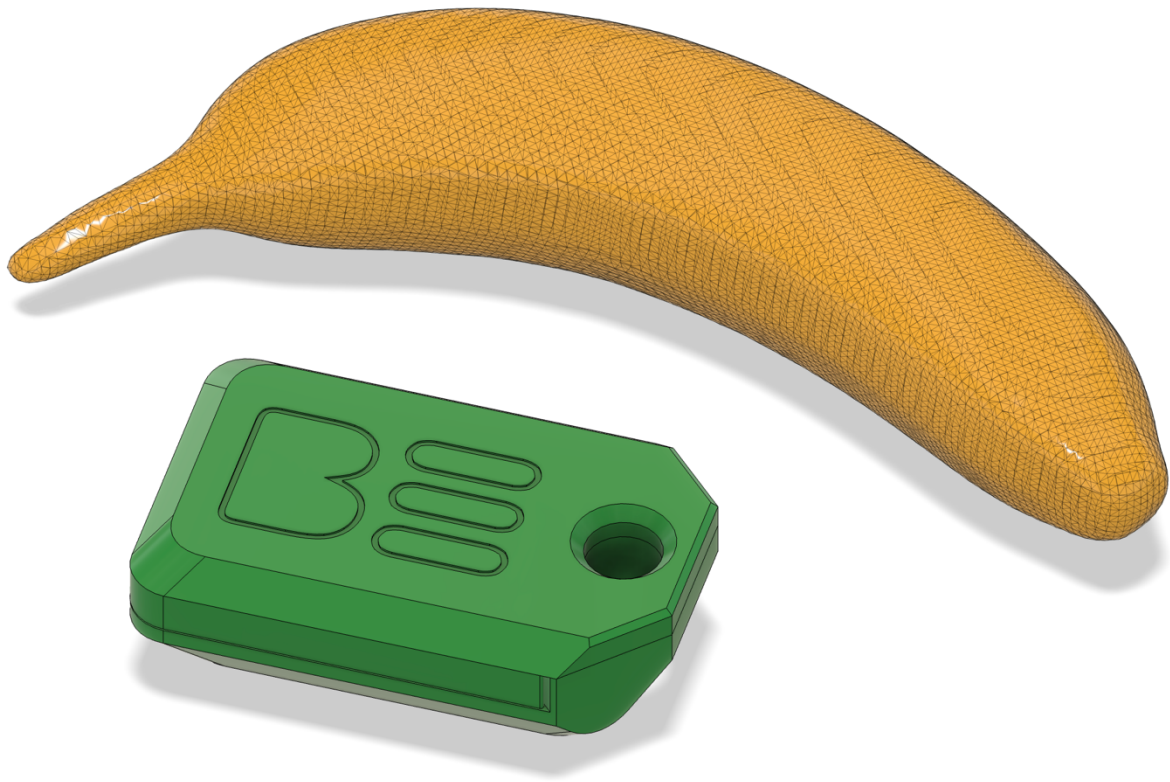




Accelerometer Holder

Memphis Zoo

dbaldrige@memphiszoo.org

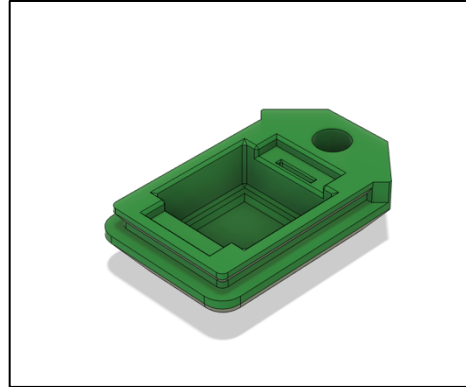
Parts:

3D Printed Parts:

Accelerometer holder lid:



Accelerometer holder body:



Stock components:

Carabiner:



[WitMotion WT901SDCL](#):



Micro SD Card:



User Guide:

Step 1:



- Plug the micro SD card into the accelerometer noting the time (preferably to the second) the blue indicator light on the front begins flashing and the sampling frequency set in the config.txt file on the card.

Step 2:



- Put the accelerometer in the housing and slide the cover closed.

Step 3:



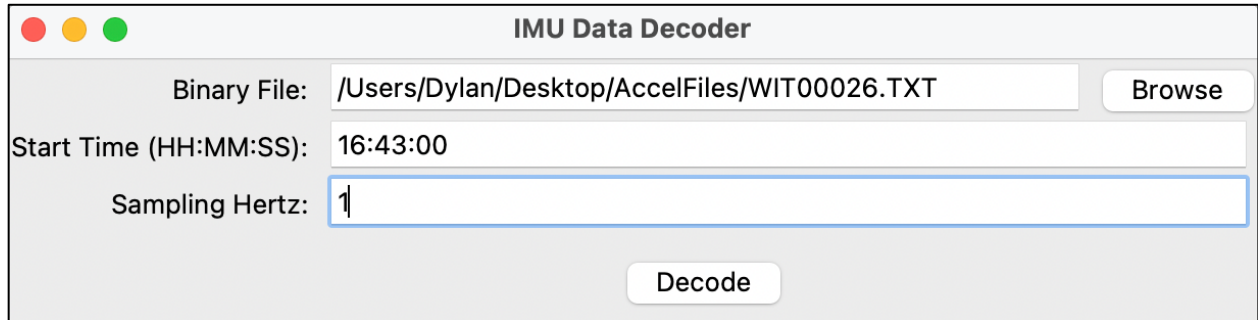
- Secure the cover closed with the carabiner and clip it to your enrichment item.

Step 4:



- Unclip your accelerometer and remove the micro SD card, turning off the logger. Note the time the SD card is removed (preferably to the second).

Step 5:



The screenshot shows a macOS-style window titled "IMU Data Decoder". It features three standard window control buttons (red, yellow, green) in the top-left corner. The interface contains three input fields: "Binary File:" with the path "/Users/Dylan/Desktop/AccelFiles/WIT00026.TXT" and a "Browse" button to its right; "Start Time (HH:MM:SS):" with the value "16:43:00"; and "Sampling Hertz:" with the value "1". A "Decode" button is positioned at the bottom center of the window.

- Using the decoder programs in the processing folder decode your data entering the file path, start time, and sampling frequency into the prompts. The exact interface will depend on the version of the decoder program works best with your computer (.app for iOS, .exe for windows, .py for both).
- *NOTE: I have not tested this on windows but wanted to include an exe for windows users. Email me if it does not work and I can help troubleshoot.*

Step 6:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Timestamp	AccX(g)	AccY(g)	AccZ(g)	GyrX(DegPer	GyrY(DegPer	GyrZ(DegPer	AngX(Deg)	AngY(Deg)	AngZ(Deg)	MagX(microt	MagY(microt	MagZ(microt
2	16:43:00	0.623	-0.125	0.797	-0.488	1.709	0.427	-8.904	-37.645	0	-273.067	0	0
3	16:43:01	0.615	-0.126	0.806	5.127	-2.258	-0.854	-8.245	-36.2	0.286	-4.633	2.033	1.667
4	16:43:02	0.573	0.032	0.826	16.907	15.198	21.118	4.373	-35.277	7.339	-4.533	1.6	1.225
5	16:43:03	0.57	0.111	0.864	12.39	-22.156	-3.052	4.625	-32.256	14.057	-4.217	1.6	1.05
6	16:43:04	0.531	-0.095	0.901	33.447	-5.371	1.892	-4.79	-29.575	20.583	-3.95	1.975	1.192
7	16:43:05	0.396	0	0.967	2.38	2.625	0.427	2.296	-22.401	18.391	-3.75	1.667	0.867
8	16:43:06	0.456	0.011	0.903	3.479	-4.089	4.578	3.571	-26.06	17.919	-3.925	1.625	0.925
9	16:43:07	0.639	0.139	0.749	50.11	-6.836	63.049	6.004	-38.738	-4.471	-5.083	1.525	1.575
10	16:43:08	0.692	0.203	0.658	30.029	3.906	4.089	13.964	-40.809	9.662	-4.767	1.317	1.142
11	16:43:09	-0.13	0.47	0.67	-11.047	-29.175	-0.488	40.765	-7.592	48.065	-1.583	-0.692	-0.567
12	16:43:10	0.005	0.007	1.038	1.587	-0.122	1.16	-8.091	-1.818	34.975	-2.2	2.233	0.325
13	16:43:11	0	0.006	1.04	1.77	-0.122	1.404	-3.082	-1.11	35.046	-2.2	2.242	0.317
14	16:43:12	0	0.006	1.041	1.77	-0.183	1.16	-0.324	-0.758	35.656	-2.2	2.242	0.325
15	16:43:13	0.001	0.007	1.038	1.526	-0.061	1.16	1.187	-0.588	36.541	-2.217	2.233	0.317
16	16:43:14	0.002	0.01	1.095	-0.244	-1.831	-0.366	1.044	-0.319	36.683	-2.217	2.242	0.308
17	16:43:15	0.005	0.004	1.041	-0.122	-0.061	0.061	0.709	-0.181	36.612	-2.2	2.233	0.317
18	16:43:16	0.001	0.005	1.037	0	-0.061	0	0.555	-0.148	36.579	-2.208	2.242	0.325
19	16:43:17	0.002	0.005	1.038	0	0	0	0.45	-0.126	36.552	-2.208	2.242	0.308
20	16:43:18	0.002	0.006	1.037	0	0	0	0.396	-0.11	36.535	-2.208	2.242	0.317
21	16:43:19	0.002	0.005	1.04	0	0	0	0.368	-0.104	36.524	-2.217	2.233	0.325
22	16:43:20	0.001	0.006	1.037	0	0	0	0.352	-0.099	36.519	-2.208	2.242	0.317
23	16:43:21	0	0.006	1.038	0	0	0	0.341	-0.093	36.519	-2.217	2.242	0.325
24	16:43:22	0	0.006	1.038	0	0	0	0.335	-0.093	36.519	-2.208	2.233	0.308
25	16:43:23	0.001	0.006	1.038	0	0	0	0.33	-0.088	36.513	-2.217	2.233	0.308
26	16:43:24	0.001	0.007	1.037	0	0	0	0.33	-0.088	36.513	-2.217	2.242	0.308
27	16:43:25	0.001	0.005	1.039	0	0	0	0.33	-0.088	36.513	-2.217	2.25	0.308
28	16:43:26	0.001	0.005	1.04	0	0	0	0.33	-0.093	36.513	-2.208	2.242	0.325
29	16:43:27	0.001	0.007	1.038	0	0	0	0.33	-0.093	36.513	-2.217	2.233	0.308
30	16:43:28	0.001	0.007	1.039	0	0	0	0.341	-0.099	36.502	-2.208	2.242	0.317
31	16:43:29	0	0.007	1.038	0	0	0	0.335	-0.093	36.497	-2.208	2.233	0.317
32	16:43:30	0.003	0.002	1.039	0	0.122	-0.122	0.335	-0.093	36.491	-2.217	2.233	0.317
33	16:43:31	0.004	0.004	1.039	0.244	0	-0.061	0.352	-0.104	36.464	-2.217	2.242	0.308
	Raw	Culled	Controls	Summary	+								

- Delete **all** data from the Raw sheet of the InterpSheet.xlsm file and paste in your decoded data (WIT#####.TXT).

Step 7:

	A	B	C
1	Trial Parameters	Start Time	11:33:15
2		Stop Time	17:11:00
3	Accelerometer Parameter	Sampling Herz	1
4	Prosscassing Parameters	Movement Magnitude Cutoff	0.5
5		Movement Axis Number Cutoff	1
6			

- Set the start time, stop time, sampling Hz for the recording on the Control sheet of the interp file.
- Set the Movement Magnitude Cutoff (°/sec) and the Movement Axis Number Cutoff (number of axes (X, Y, and Z) that have to meet this magnitude for a reading to count as engagement.

Step 8:

	A	B	C	
1	Activation	Seconds	Minutes	
2	X Activation	61.00	1.02	
3	Y Activation	59.00	0.98	
4	Z Activation	57.00	0.95	
5	Mean Activation	59.00	0.98	
6	Combined Activation	65.00	1.08	
7				
8				
9	250			
10				
RawCulledControlsSummary+				

- Look at the summary table and graph for the entered parameters and data on the Summary sheet.
 - o Mean activation: time activation according to the Movement Magnitude Cutoff parameter in the control sheet was recorded averaged across all three axes
 - o Combined activation: time activation was recorded according to the Movement Axis Number Cutoff and Movement Magnitude Cutoff parameters on the Control Sheet.